

Data Analytics Capstone

Forecasting Short-Term Exchange Rate Movements and Volatility in Major Currency Pairs Using Macroeconomic Indicators and Machine Learning

Synopsis

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QM640: Data Analytics Capstone

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Fall 2026 Term

August 22, 2026

Introduction

Background and Context

The foreign exchange (FX) market is the largest and most liquid financial market in the world, with global turnover exceeding USD 7 trillion per day (Bank for International Settlements, 2022). Currency values fluctuate continuously in response to macroeconomic fundamentals, central bank policy actions, and shifting investor sentiment, creating both risk and opportunity for the institutions and individuals who transact across borders. For corporate treasuries, importers and exporters, institutional investors, and retail traders alike, the ability to anticipate short-term currency movements directly affects hedging decisions, pricing strategy, and portfolio performance.

Exchange rates are theoretically linked to macroeconomic fundamentals through relationships such as uncovered interest rate parity (which ties currency movements to interest-rate differentials) and purchasing power parity (which ties currency movements to inflation differentials). In practice, however, these fundamentals explain only part of short-term currency behavior. Meese and Rogoff (1983) famously demonstrated that structural macroeconomic models often fail to outperform a naive random-walk forecast over short horizons — a finding often referred to as the “Meese-Rogoff puzzle” — which has motivated decades of subsequent research into alternative forecasting approaches.

In recent years, machine learning (ML) methods such as random forests, gradient-boosted trees (XGBoost), and long short-term memory (LSTM) networks have shown an ability to capture nonlinear and interaction effects that classical econometric models such as ARIMA and GARCH may miss (Bollerslev, 1986; Breiman, 2001; Chen & Guestrin, 2016; Hochreiter & Schmidhuber, 1997). Separately, natural language processing (NLP) techniques now make it possible to quantify sentiment embedded in financial news and social media in near real time (Araci, 2019; Hutto & Gilbert, 2014), and research has linked such sentiment measures to

subsequent market movements (Tetlock, 2007). Central bank policy announcements are a further well-documented source of short-term volatility (Kuttner, 2001), particularly for emerging-market currencies such as the Indian rupee, which are highly sensitive to shifts in U.S. monetary policy and global risk appetite.

This project sits at the intersection of these strands of research. It is directly relevant to corporate finance and treasury teams that must decide when and how to hedge currency exposure, to FX traders and portfolio managers who require timely directional and volatility signals, and to policymakers monitoring exchange-rate stability. By systematically comparing macroeconomic, machine-learning, sentiment-augmented, and event-based approaches on a common set of currency pairs, this study aims to generate evidence that is directly actionable for these stakeholders.

Problem Statement

Despite decades of research, forecasting short-term exchange-rate direction and volatility remains difficult, and existing studies typically evaluate macroeconomic-fundamental models, machine-learning models, sentiment-based signals, and policy-event effects in isolation rather than within a single, comparable framework. This fragmentation leaves practitioners without clear, comparative evidence on which class of model — or which combination of signals — delivers the most reliable short-term forecasts for major currency pairs, and specifically for an emerging-market currency (the Indian rupee) alongside major developed-market pairs. This project addresses that gap by evaluating, on the same currency pairs and time horizon, whether machine-learning models outperform traditional econometric benchmarks, whether financial-news sentiment adds incremental predictive value, and how central bank policy announcements measurably affect short-term volatility.

Purpose of the Study

The purpose of this study is to compare the effectiveness of macroeconomic-indicator-based models and machine-learning models in predicting the short-term (daily) directional movement and volatility of four major currency pairs — USD/INR, EUR/USD, GBP/USD, and USD/JPY — and to evaluate whether financial-news sentiment and central-bank policy events materially improve forecast accuracy. The study is both predictive (forecasting direction and volatility) and explanatory (identifying which macroeconomic, sentiment, and policy drivers are statistically significant), and it draws on regression, classification, time-series econometrics, and NLP-based sentiment analysis.

Each objective below is written to be Specific, Measurable, Achievable, Relevant, and Time-bound (SMART): (1) develop a directional-accuracy forecasting model for four currency pairs using macroeconomic predictors by Week 5; (2) benchmark machine-learning models against ARIMA/GARCH baselines and demonstrate a measurable accuracy improvement by Week 6; (3) quantify the incremental accuracy gain from adding news sentiment by Week 7; and (4) establish a statistically significant link between policy surprises and volatility spikes by Week 8 — each objective is achievable using only public data and standard statistical/ML tooling within the 10-week capstone window, and is directly relevant to currency-risk decision-making in corporate treasury and trading contexts.

Scope and Objectives

The research problem is broken down into four research questions, each mapped to a specific analytic method and data requirement, so that the overall scope remains broad enough to yield meaningful, generalizable insight while staying feasible within a 10-week capstone timeline.

Research Question 1 (RQ1)

To what extent do macroeconomic indicators — interest-rate differential, inflation differential, trade balance, and GDP growth rate — predict the short-term directional movement of major currency pairs (USD/INR, EUR/USD, GBP/USD, USD/JPY)?

H0: None of the macroeconomic predictors is significantly associated with short-term directional currency movement. *Ha*: At least one macroeconomic predictor — most likely the interest-rate or inflation differential — is significantly associated with directional movement.

Research Question 2 (RQ2)

Do machine-learning models (Random Forest, XGBoost, and LSTM) provide significantly better forecasting accuracy for exchange-rate direction and volatility than traditional econometric models (ARIMA and GARCH)?

H0: The directional/volatility forecast accuracy of the ML models is not significantly different from that of the ARIMA/GARCH baselines. *Ha*: The ML models achieve significantly higher accuracy than the econometric baselines, with an improvement of at least 10 percentage points.

Research Question 3 (RQ3)

Does incorporating financial-news sentiment, extracted using NLP sentiment-scoring methods, significantly improve the predictive accuracy of exchange-rate direction models beyond macroeconomic indicators alone?

H0: Adding the news-sentiment feature does not significantly improve directional accuracy over the best RQ2 model. *Ha*: Adding the news-sentiment feature significantly improves directional accuracy, by at least 5 percentage points.

Research Question 4 (RQ4)

Is there a statistically significant relationship between central bank policy announcements (Federal Reserve, RBI, ECB, and Bank of England interest-rate decisions) and short-term spikes in exchange-rate volatility, and can the size of a policy “surprise” predict the magnitude of that spike?

H0: Realized volatility immediately after a policy announcement is not significantly different from the pre-announcement baseline, and policy surprises do not predict spike magnitude. *Ha*: Post-announcement volatility is significantly higher than the pre-announcement baseline, and larger policy surprises predict larger spikes.

Key Variables

The table below summarizes the primary variables used across the four research questions, including their type, unit of measurement, analytic role, and data source, to ensure that every construct in the study is clearly measurable.

Variable	Type	Unit	Role	RQ	Source
Exchange rate (close)	Continuous	Local currency per USD	Base series	RQ1–RQ4	RBI DBIE / ECB SDW / BoE / MoF Japan
FX daily return	Continuous	% log return	Outcome	RQ1, RQ2	Derived
Realized volatility	Continuous	Annualized %	Outcome	RQ2, RQ4	Derived (rolling std. dev.)
Interest-rate differential	Continuous	Percentage points	Predictor	RQ1	FRED / RBI / ECB / BoE
Inflation differential	Continuous	Percentage points	Predictor	RQ1	FRED / IMF IFS
Trade balance	Continuous	USD millions	Predictor	RQ1	IMF IFS / national statistics agencies
GDP growth rate	Continuous	% YoY	Predictor	RQ1	World Bank / IMF IFS
News sentiment score	Continuous	Compound score, -1 to +1	Predictor	RQ3	GDELT / news headlines (VADER, FinBERT)
CB announcement flag	Binary	0 / 1	Predictor	RQ4	Fed / RBI / ECB / BoE calendars
Policy surprise	Continuous	Basis points	Predictor	RQ4	Actual vs. consensus-expected decision
Currency pair	Categorical	4 levels	Grouping	RQ1–RQ4	—
Date	Date/time	Daily	Time index	RQ1–RQ4	—

Data Availability and Ethical Considerations

All data required for this study are publicly available from official government, central-bank, and international-organization sources (e.g., FRED, RBI DBIE, ECB SDW, Bank of England, IMF IFS, World Bank, and the GDELT Project); no Kaggle-hosted datasets are used, consistent with course requirements. Because the data consist entirely of aggregate macroeconomic series, market prices, and publicly published news content, no personally identifiable information is collected, and no human-subject consent procedures are required. Two limitations are nonetheless noted: first, macroeconomic series such as GDP and trade balance are periodically revised after initial release, so this study will use the most recently available vintage and flag this as a limitation rather than attempt full real-time reproducibility; second, sentiment will be computed only from news published strictly before each prediction date to avoid look-ahead bias.

Feasibility and Project Timeline (10 Weeks)

- Week 1: Finalize problem statement and research questions; complete literature review; obtain mentor approval of the synopsis.
- Week 2: Collect historical FX rates and macroeconomic indicators (interest rates, inflation, trade balance, GDP growth) from FRED, RBI DBIE, ECB SDW, and IMF IFS.
- Week 3: Collect central bank announcement calendars and GDELT news data; clean and align data across daily, monthly, and event-based frequencies.
- Week 4: Conduct exploratory data analysis, stationarity testing, and feature engineering (e.g., interest-rate and inflation differentials, rolling volatility).
- Week 5: Build and validate the RQ1 regression models; first mentor checkpoint / interim report.

- Week 6: Build ARIMA/GARCH baselines and Random Forest/XGBoost/LSTM models for RQ2; run Diebold-Mariano comparison tests.
- Week 7: Extract NLP sentiment scores and augment the best RQ2 model for RQ3; compare accuracy with and without sentiment.
- Week 8: Conduct the RQ4 event-study analysis around central bank announcements; second mentor checkpoint.
- Week 9: Synthesize findings across all four research questions; draft recommendations and visualizations.
- Week 10: Finalize the report, code repository, and final presentation; final mentor review.

This phased plan keeps each research question scoped to roughly one to two weeks of modeling work, uses exclusively pre-existing public data (avoiding primary data-collection delays), and reuses a common data-preparation pipeline across all four research questions, making the overall scope realistic within the 10-week window.

Sample Size Calculation

Each research question uses the method most appropriate to its analytic design — multiple regression (Green’s rule), a two-proportion power calculation (for comparing model accuracies), or a correlation-based power formula (for the event-study relationship) — as summarized below. The two-proportion calculations follow the classical power-analysis principle that, for a 95% confidence level and 80% power, the hypothesized effect must be at least 2.8 standard errors from the null comparison point (1.96 for the 95% interval plus 0.84 to reach the 80th percentile of the normal distribution); this yields the general form $n = [2.8 / (p_1 - p_2)]^2 \cdot [p_1(1 - p_1) + p_2(1 - p_2)]$, consistent with Gelman and Hill (2007, Ch. 20).

RQ	Method	Key Parameters	Min. N	Unit
RQ1	Multiple Regression (Green’s Rule)	$k = 5$ predictors (interest-rate diff., inflation diff., trade balance, GDP growth, lagged return); $n \geq 50 + 8k$	$n \geq 90$	Monthly, per pair

RQ	Method	Key Parameters	Min. N	Unit
RQ2	Two-proportion power test	$\alpha = 0.05$, Power = 0.80, p_1 (econometric) = 0.55, p_2 (ML) = 0.65	$n \geq 373$ /grp	Daily forecast days
RQ3	Two-proportion power test	$\alpha = 0.05$, Power = 0.80, $p_1 = 0.65$, $p_2 = 0.72$	$n \geq 687$ /grp	Daily forecast days
RQ4	Correlation-based power formula	$\alpha = 0.05$, Power = 0.80, expected $r = 0.30$; $n = [(Z(1-\alpha/2) + Z(1-\beta)) / (0.5 \cdot \ln((1+r)/(1-r)))]^2 + 3$	$n \geq 85$ events	Per CB policy event

To ensure adequate statistical power across all analyses, the final sample size is selected as the maximum of the minimum sample sizes obtained across RQ1–RQ4, which is $n \geq 687$ daily observations (driven by RQ3). To provide a comfortable margin for model training, validation, and testing splits — and to allow the RQ4 event study to draw on a sufficient number of policy-announcement dates — this study will collect approximately eight years of daily data (2018–2026, roughly 2,000+ trading days) per currency pair, which comfortably exceeds every minimum threshold above.

Data Description

Data Sources

- Federal Reserve Economic Data (FRED) — U.S. interest rates, CPI inflation: fred.stlouisfed.org
- RBI Database on Indian Economy (DBIE) — India policy rate, inflation, USD/INR rates: dbie.rbi.org.in
- European Central Bank Statistical Data Warehouse (SDW) — Eurozone rates, EUR/USD: sdw.ecb.europa.eu
- Bank of England Statistical Interactive Database — UK rates, GBP/USD: bankofengland.co.uk/statistics
- IMF International Financial Statistics (IFS) — cross-country trade balance and GDP data: data.imf.org
- World Bank Open Data — GDP growth rates: data.worldbank.org

- GDELT Project — public news tone/sentiment data (non-Kaggle): gdeltproject.org
- Official policy calendars — Federal Reserve (FOMC), RBI MPC, ECB Governing Council, and Bank of England MPC meeting schedules and decisions.

Data Dictionary

Dataset	Frequency / Coverage	Description
Historical FX rates (USD/INR, EUR/USD, GBP/USD, USD/JPY)	Daily, 2018–2026	Official closing rates published by RBI DBIE, ECB SDW, Bank of England, and Bank of Japan / Ministry of Finance Japan.
Interest rates	Daily/monthly, 2018–2026	Policy rates from FRED (Fed Funds), RBI (repo rate), ECB (MRO rate), BoE (Bank Rate).
Inflation (CPI)	Monthly, 2018–2026	Consumer price index, year-over-year, from FRED and IMF International Financial Statistics.
Trade balance & GDP growth	Monthly / quarterly	From IMF IFS and World Bank national accounts data.
Financial news headlines	Daily, 2018–2026	Tone and event data from the GDELT Project (public, free, non-Kaggle); used to compute sentiment scores.
Central-bank policy calendars	Event-based, 2018–2026	Official FOMC, RBI MPC, ECB Governing Council, and BoE MPC meeting dates and decisions.

Code and Data Repository

All collected data, cleaning scripts, and modeling code will be maintained in a public GitHub repository (link: <GitHub repository URL to be added>), organized into clearly labeled folders for raw data, processed data, notebooks, and results, with a top-level README describing the folder structure and reproduction steps, consistent with course requirements for open access to data and code.

Analytic Approach

Solution to RQ1

RQ1 will be addressed using multiple linear regression (OLS) of daily/monthly FX returns on the macroeconomic predictors listed above, with currency pair included as a fixed effect. Before interpreting coefficients, each predictor will be checked for multicollinearity using the

Variance Inflation Factor (VIF), with VIF above 10 flagged as significant and addressed by dropping or combining correlated predictors. As a nonlinear robustness check, a Random Forest regression will be fit on the same predictors and validated using 5-fold cross-validation to guard against overfitting. Hypothesis: interest-rate and inflation differentials will show the strongest association with directional currency movement, consistent with uncovered interest-rate parity and purchasing-power-parity theory. Model outputs will be evaluated using R^2 , adjusted R^2 (to penalize added predictors fairly), RMSE, and MAE, along with residual diagnostics for heteroskedasticity and autocorrelation.

Solution to RQ2

RQ2 will compare ARIMA and GARCH models (statistical baselines capturing volatility clustering) against Random Forest, XGBoost, and LSTM models (machine-learning approaches) for forecasting next-day return direction and volatility. ARIMA/GARCH order will be selected using AIC/BIC model-selection criteria, and all models will be trained on a rolling-window basis with time-series cross-validation (no random shuffling, to avoid look-ahead leakage) and evaluated on a held-out test period. Hypothesis: ML models will achieve higher directional accuracy and lower volatility-forecast error than the econometric baselines, particularly during high-volatility regimes. Accuracy will be assessed via a confusion matrix (true/false positives and negatives) and compared using RMSE, MAPE, directional accuracy, and the Diebold-Mariano test to assess whether accuracy differences are statistically significant (Diebold & Mariano, 1995).

Solution to RQ3

RQ3 will extract sentiment scores from financial news headlines and GDELT tone data using rule-based (VADER) and transformer-based (FinBERT) sentiment models (Araci, 2019; Hutto & Gilbert, 2014), aggregated to a daily sentiment index per currency pair. This sentiment feature will be added to the best-performing model identified in RQ2, and the

augmented model's accuracy will be compared to the non-augmented model using McNemar's test on paired directional predictions (i.e., comparing confusion-matrix outcomes for the same test days under each model). Hypothesis: incorporating sentiment will yield a statistically significant, though modest, improvement in directional accuracy, consistent with prior findings linking media sentiment to short-term market moves (Tetlock, 2007).

Solution to RQ4

RQ4 will use an event-study design: for each central bank policy announcement date, realized volatility (derived from a GARCH conditional-variance estimate) in the window immediately following the announcement will be compared to a matched pre-announcement window using a paired t-test or, if normality assumptions are violated, a Wilcoxon signed-rank test. A logistic regression will further model the probability of a volatility spike as a function of the "policy surprise" (the gap between the actual and consensus-expected rate decision), with odds ratios used to interpret the effect size and model fit assessed via AIC. Hypothesis: larger policy surprises will be associated with a significantly higher probability and magnitude of short-term volatility spikes (Kuttner, 2001).

Evaluation Metrics

- Regression tasks (RQ1): R^2 , Adjusted R^2 , $RMSE = \sqrt{\sum(y_{\text{pred}} - y)^2 / n}$, $MAE = \sum|y_{\text{pred}} - y| / n$.
- Forecasting tasks (RQ2, RQ3): $MAPE = (100/n) \cdot \sum|(y - y_{\text{pred}})/y|$, directional accuracy = correct direction calls / total calls, and the Diebold-Mariano test statistic for comparing forecast errors between model pairs.
- Classification comparison (RQ2, RQ3): confusion-matrix-based Accuracy, Precision, Recall, F1-score, and AUC-ROC; McNemar's chi-square test for paired-model accuracy comparisons.

- Event study (RQ4): paired t-test / Wilcoxon signed-rank statistic for pre- vs. post-announcement volatility, and pseudo- R^2 / odds ratios for the logistic regression.

Recommendation and Application

The intended end users of this research are corporate treasury and finance teams managing currency exposure, FX traders and portfolio risk managers, and import/export businesses that need short-term guidance on currency risk. Based on the anticipated findings, the study will recommend which model family (macroeconomic regression, ML, sentiment-augmented, or event-based) is most reliable for each type of decision — for example, using macroeconomic regression for medium-term hedging decisions, ML/sentiment-augmented models for short-term tactical trading signals, and the event-study results to time hedges around known central-bank policy dates. A practical deliverable will be a simple dashboard-style summary (built from the model outputs) that flags elevated volatility risk ahead of scheduled policy announcements, giving treasury teams an actionable early-warning signal rather than a purely academic forecast.

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